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Professional Summary

AI Engineer and GenAI Developer specializing in **Autonomous AI Agents**, **LLM Applications**, **RAG systems**, **Vector Search**, and **intelligent automation**. Skilled in building production-grade agentic workflows using **LangGraph**, **LangChain**, **GPT/Gemini/Claude models**. Proven record of **40% accuracy improvement** and **30% manual effort reduction** in enterprise workflows.

Core Skills

AI & GenAI: LLM Applications, Generative AI, Prompt Engineering, Embeddings, Tool Calling

Agentic AI: LangGraph, Google ADK, LangChain, MCP, Multi-Agent Systems, Agent Orchestration

RAG & Retrieval: RAG, Hybrid Search, Semantic Search, FAISS, BM25, Vector Databases, Retrieval Optimization

Evaluation & Testing: DeepEval, RAMPART, Agent Evaluation, LLM Evaluation, Safety Testing, Prompt Injection Testing, pytest

AI Safety: Guardrails, Validation Layers, Hallucination Mitigation, Human-in-the-Loop

Machine Learning: Scikit-learn, TensorFlow, PyTorch, NumPy, Pandas

Backend & APIs: Python, FastAPI, REST APIs, PostgreSQL, SQLite

Data & Automation: Playwright, Web Scraping, Async Python, Data Extraction

MCP & Integrations: Model Context Protocol (MCP), Trino MCP, Tool Integration

Tools & Deployment: Git, Docker, AWS, CI/CD

Education

M.E. Computer Science

Kings Engineering College 2025–Present

B.Tech Information Technology

DMI College of Engineering 2021–2025
CGPA: 8.4/10

Certifications

- Claude Certified Architect – Foundations — **Anthropic**
- Model Context Protocol: Advanced Topics — **Anthropic**
- Hugging Face Agents Course — **Hugging Face**
- ACP: Agent Communication Protocol — **DeepLearning.AI**
- Agentic Knowledge Graph Construction — **DeepLearning.AI**
- Machine Learning with Python — **IBM**

Achievements

- 1st — National Hackathon (Alpha College)
- 3rd — SheInnovates Hackathon (WTM)
- 2nd — National Coding Contest (CIT)

Languages

Tamil: Native Proficiency

English: Professional Working Proficiency

Experience

Associate Data Analyst

Sep 2025 – Present

Zinnov (Draup)

Nemili, India

- Architected **DRI-MAS**, a **LangGraph-based multi-agent system** to automate enterprise data requests, transforming natural-language queries into validated **Trino Mcp queries** and **structured Excel reports**, reducing analyst turnaround time from hours to **under 1 minute**.
- Built **10+ production components** for agent orchestration, including intent routing, normalization, schema selection, SQL generation, memory, evaluation, and **FastAPI** services, improving reliability of AI-driven data workflows.
- Implemented **FAISS + BM25 hybrid retrieval** across **2.6M company records** to improve enterprise semantic search and support Human-in-the-Loop validation workflows.
- Developed a **Company Synonym Agent** for entity resolution across **2.4M+ company and subsidiary records**, combining semantic search, BM25, acronym validation, and parent-company resolution to achieve **sub-second retrieval**.
- Automated business-signal monitoring through the **DOF Agent** for **537 enterprise accounts across 8 industries**, integrating web scraping, deduplication, LLM-based classification, scoring, and automated reporting workflows.
- Developed AI-powered workforce intelligence solutions, including **746 role-task-skill mappings**, **30+ automated test suites**, **4 production APIs**, and an **AI-assisted Proficiency Model** using structured evidence, confidence scoring, and Human-in-the-Loop validation.

Technologies: Python, LangGraph, Claude API, FastAPI, Trino Mcp, PostgreSQL, FAISS, BM25, AWS S3, AWS Glue, Playwright, Pandas, Pydantic, pytest

Tech Intern

Jun 2025 – Sep 2025

Zinnov (Draup)

Nemili, India

- Developed the **RMS Agent**, an AI-powered role mapping system that standardized job descriptions against Draup's workforce taxonomy using automated classification workflows.
- Built **LLM-driven job description enrichment and skill extraction pipelines** using **Gemini 2.0 Flash** and **FAISS semantic search**, supporting scalable role classification and skills intelligence.
- Automated identification of **Existing, New, and Intern Roles** through parallel batch-processing pipelines, improving scalability and consistency of workforce data processing.
- Built foundational business-signal intelligence pipelines using **Requests, BeautifulSoup, Playwright, and GNews APIs**, incorporating validation, deduplication, and a documented **6-stage automation architecture** later expanded into enterprise-scale workflows.

Technologies: Python, FAISS, Gemini 2.0 Flash, Playwright, BeautifulSoup, Requests, GNews API, Pandas, Mermaid, OpenPyXL

Key Projects

- **Finn Advisor** — AI financial advisor built with **Python, Google ADK, LangGraph, MCP, LLMs, and SQLite**. Developed **agent orchestration**, **tool calling**, **RAG-style web research**, and **stateful workflows** for portfolio analysis using Kite MCP. Implemented **multi-provider LLM routing**, **rate limiting**, **fallback handling**, **memory**, **observability**, and **safety evaluation** with NVIDIA NIM, Gemini Flash, DeepEval, and RAMPART.
- **Instagram Alpic MCP Server Live & Deployed** — Production Python MCP server with Composio OAuth, Instagram Graph API, real-time messaging and analytics, monitoring, and Alpic deployment.
- **LinkedIn Alpic MCP Server Live & Deployed** — Production MCP server for real-time posting, messaging, analytics, and profiles using LinkedIn APIs, OAuth 2.0, and Composio.
- **LinkedIn MCP Server** — Browser-automation and Graph API-based MCP tool for profile, company, job, and post workflows with modular AI-agent integration.
- **Outlook CLI** — Stdio-, HTTP-, and CLI-powered MCP server for Outlook email, calendar, contacts, and automation using Microsoft Graph and OAuth 2.0.
- **ChunkLab** — Interactive RAG platform comparing fixed, recursive, semantic, and structure-aware chunking using FAISS, Sentence Transformers, Recall@K, MRR, Gradio, CLI, and HTML/JSON reports.
- **Company Logo Finder** — Python multi-source logo retrieval pipeline with intelligent company matching and resilient fallback strategies.
- **AI Calculator** — Responsive handwritten-equation solver built with Next.js, Gemini AI, Canvas API, TypeScript, and Tailwind CSS.